

Optimization of Load Balancing in Hybrid LiFi/RF Networks

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Abstract—Light fidelity (LiFi) uses light emitting diodes (LEDs) for high-speed wireless communications. Since an LED lamp covers a small area, a LiFi system with multiple access points (APs) can offer a significantly high spatial throughput. However, the spatial distribution of data rates achieved by LiFi fluctuates because users experience inter-cell interference from neighboring LiFi APs. In order to guarantee a quality of service (QoS) for all users in the network, an RF network is considered as an additional wireless networking layer. This hybrid LiFi/RF network enables users with low levels of optical signals to achieve the desired QoS by migrating to the RF network. With regard to moving users, the hybrid LiFi/RF system dynamically allocates either a LiFi AP or an RF AP to users based on their channel state information. In this paper, a dynamic load balancing scheme is proposed, which considers the handover overhead in order to improve the overall system throughput. Joint optimization algorithm (JOA) and separate optimization algorithm (SOA), which jointly and separately optimize the AP assignment and resource allocation, respectively, are proposed. Simulation results show that SOA can offer a better performance/complexity tradeoff than JOA for system load balancing.

Index Terms—LiFi, RF, dynamic load balancing, handover overhead, outage.

I. INTRODUCTION

THE radio frequency (RF) spectrum exhibits a looming exhaustion due to the exponentially increasing demand for wireless and mobile data communications. Due to more than 300 THz visible light range of the electromagnetic spectrum, which is 600,000 times more than a 500 MHz WiGig (wireless gigabit alliance) channel, Light Fidelity (LiFi) technology has been proposed as a potential solution [1]. According to [2], a data rate of 3 Gb/s is achieved by a single colour light emitting diode (LED). Moreover, latest research shows that 100 Gb/s optical wireless data rate is achievable at standard indoor illumination levels by using off-the-shelf laser diodes (LDs) [3].

In indoor environments, a single LiFi cell covers a few square meters due to propagation characteristics of light such

as high path loss and low multipath reflections. Hence, multiple light sources are required to cover a large room, and because of this effect high spatial spectral efficiency can be achieved by LiFi. In spite of the dense deployment of access points (APs), LiFi may not provide a uniform coverage in terms of data rate performance mainly due to inter-cell interference (ICI) [4]. Moreover, if the line-of-sight (LoS) optical channels are blocked by obstacles, the performance of per user data rate would be further compromised due to the low optical channel gain. It has been shown that although LiFi networks can provide significantly high system data rates, the outage probability of per user data rate in a multi-user environment can still be high [5]. It has been further shown in [5] that a well designed hybrid LiFi and RF network is able to improve both the average data rate and the outage performance. Since RF and LiFi use non-overlapping spectrum, a hybrid network will offer a total system throughput greater than that of a stand alone LiFi or RF network. The conventional indoor RF systems provide ubiquitous coverage and ensure moderate throughput at all locations. Thus, the hybrid LiFi/RF network can harness the benefits of both system throughput and user experience, which makes it a promising candidate for high-speed indoor wireless communications.

Effective load balancing for hybrid networks is a technical challenge, and some research on this topic has been undertaken [6]–[9]. *Rahaim et al.* pioneered the early research on visible light communication (VLC)/RF hybrid network on the topic of system throughput optimisation [6]. An experimental study of a practical hybrid VLC/WiFi (Wireless Fidelity) system was given in [7]. The authors implemented an asymmetric system comprised of a WiFi uplink and a VLC downlink, which differs from the network structure of LiFi/RF downlink combination. The majority of subsequent research focused on system load balancing so as to improve the performance of data rate and user fairness [8], [9]. However, most research has not considered the handover overhead. In contrast to the outdoor heterogeneous networks, the cell size of indoor LiFi/RF hybrid networks is very small so that the movement of users may frequently prompt handovers. During a handover, the signalling information is exchanged between users and the central unit (CU). This process takes time ranging from around 30 ms to 3000 ms on average, depending on the algorithm used [10]–[12]. In general, load balancing entails AP assignment (APA) and resource allocation (RA), and both steps are under the influence of the handover overhead. In practice, the channel state information (CSI) of LiFi and RF systems is time-varying because of the user movement,

Manuscript received April 5, 2016; revised August 27, 2016, November 3, 2016, and January 5, 2017; accepted January 6, 2017. Date of publication January 17, 2017; date of current version April 14, 2017. This publication was made possible by the EPSRC under Established Career Fellowship Grant EP/K008757/1. The associate editor coordinating the review of this paper and approving it for publication was W. C. Kwong.

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Digital Object Identifier 10.1109/TCOMM.2017.2654249

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and this dynamic system can be divided into many quasi-static periods with a short duration which is referred to as a ‘state’. In our previous work, a dynamic load balancing algorithm considering proportional fairness among the user population was proposed. Moreover, the handover boundary and the relationship between LiFi throughput and RF throughput were analysed [13]. In this paper, a comprehensive study of dynamic load balancing aiming at improving user data rates is undertaken, where a variety of fairness schemes and user quality of service (QoS) requirements are taken into account.

A. Main Contributions

- A novel dynamic load balancing scheme is proposed and the β -proportional fairness function is applied, where β is the fairness coefficient. The effect of β on user data rate performance in the hybrid network is evaluated.
- Two specific algorithms that optimise the APA and the RA in each quasi-static state are proposed, termed as joint optimisation algorithm (JOA) and separate optimisation algorithm (SOA) respectively. In this work, the optimality of JOA and the optimal threshold in SOA are analysed. Also, a comparison of data rate performance and computational complexity between these two algorithms is made.
- A unified data rate requirement of users is considered as a QoS metric and user outage probability is introduced. In this paper, an optimisation problem that maximises the achievable user QoS with certain outage probability is formulated and solved by numerical simulations. The maximal user QoS in the dynamic hybrid system is evaluated for both JOA and SOA.

The rest of the paper is organised as follows. The mobile hybrid LiFi and RF system model is introduced in Section II. The dynamic load balancing scheme with handover is given in Section III. Two specific load balancing algorithms, JOA and SOA, are presented in Section IV and Section V, respectively. The QoS enhancement in JOA and SOA is given in Section VI. The performance evaluation and discussion are presented in Section VII, and the conclusion is given in Section VIII.

II. SYSTEM MODEL

A. System Setup

Referring to Fig. 1, a multi-user indoor hybrid LiFi/RF network is considered, where N_l LiFi APs and N_r RF APs are deployed. It is assumed that users are distributed uniformly over the coverage area. All LiFi and RF APs are connected to a central unit (CU) through error free inter-connection links, which can be either optical fibres or power-line backbone. Each LiFi AP is a large light emitting diode (LED) lamp consisting of several low power LEDs, and each user has a photo detector (PD), assumed to be oriented perpendicular to the floor. This means that the angle of irradiance is equal to the angle of incidence in the LoS optical channel. The field of view (FoV) of the LEDs is restricted so that the communication is confined within a certain area. Thus each LiFi AP covers a small cell which is referred to as an optical attocell. In addition, all LiFi APs reuse the same modulation

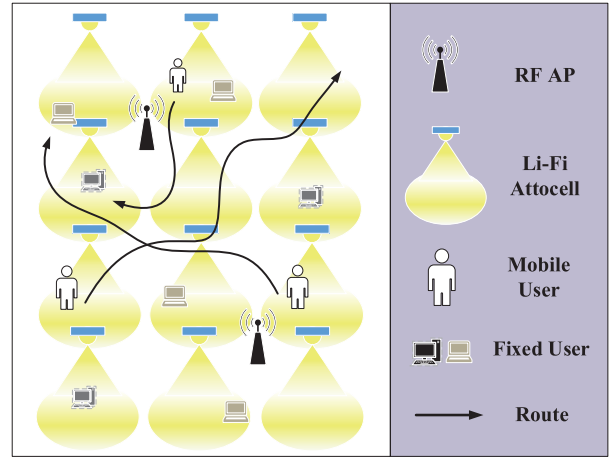


Fig. 1. Schematic diagram of the system model.

bandwidth, and users residing in the cell overlapping areas may experience optical ICI, which is treated as additional noise [13]. The spectrum reuse enables the LiFi system to provide high spatial spectral efficiency to users [14]. However, due to severe ICI, the user data rate can significantly be low over the coverage area [5]. In order to improve the per user data rate performance, the system is augmented by a RF network. The RF APs are assumed to cover the entire indoor area, and each RF AP is allocated a non-overlapping RF spectrum to avoid ICI in the RF system.

In this hybrid network, time division multiple access (TDMA) is used in both LiFi and RF systems. The network load balancing consists of AP assignment and time slot resource allocation. Due to the varying channels in dynamic indoor environments, the number of users assigned to each cell and the time resource for each user should be changed as CSI changes. However, it is important to note that unlike in RF, channels in a LiFi system do not exhibit fading characteristics as the detector size is much larger than the wavelength. They only exhibit shadowing effects. This has important implications on the channel estimation which is beyond the scope of this work. In this study, a slow walking scenario with an average speed of 0.5 m/s is considered, where the CSI in both LiFi and RF systems is assumed to be constant for a very short period, and such a period is referred to as a quasi-static state [15]. Moreover, the wavelength division duplex (WDD) method is employed in the LiFi system where the infra-red and the visible light spectra are used for uplink and downlink, respectively. The LiFi downlink channel estimation can be undertaken at the end of each quasi-static state, and the CU receives the CSI feedback via a 10 Mbps infra-red uplink, shown in [16]. The typical size of CSI feedback packets is several kilobits [17]. Therefore, the feedback delay can be less than 1 millisecond which is considered negligible in this study. The estimated downlink CSI in the LiFi system is considered to be valid for the following state. In the RF system, the time division duplex (TDD) method is used and the downlink CSI could be obtained based on the channel reciprocity, but this is beyond the scope of this work.

The time interval of each state is denoted by T_p (second). The CU monitors the system continuously in every such an interval, where the load distribution is assumed to be fixed and users receive a constant data rate. An integer, n , is defined as the sequence number of the state. Handover occurs when the serving AP of a user changes as users move. In this study, the overhead, and in turn the reduction of per user data rate are considered in the performance evaluation.

The set of users is denoted by $\mathcal{U}$; the number of the users is denoted by N_u ; the set of optical attocells is denoted by $\mathcal{C}_L = \{l | l \in [1, N_l], l \in \mathbb{N}\}$; and $\mathcal{C}_R = \{r | r \in [1, N_r], r \in \mathbb{N}\}$ is the set of RF cells.

B. LiFi Channel Model

LiFi uses baseband communication which relies on intensity modulation (IM) at the transmitter and the direct detection (DD) at the receiver. Hence, only positive real-valued signals can be transmitted [18]. In this study, direct current biased optical orthogonal frequency division multiplexing (DCO-OFDM) is used [18]. The direct current (DC) bias added to the modulated signal ensures that the output signals are positive. Also, at least half of the OFDM sub-carriers are used to realise the Hermitian conjugate of the complex-valued symbols so that the strict positive and real number constraint is satisfied after the Fourier transform. The LiFi channel coefficient at the OFDM sub-carrier frequency, f , is given by [19]:

$$H_{\mu,\alpha}(f) = H_C(f)H_F(f), \quad (1)$$

where $H_C(f)$ is the optical channel gain; and $H_F(f)$ is the front-end device frequency response. Here front-end device means both the LED at the transmitter and the PD at the receiver. According to [19], the optical channel gain can be modelled by:

$$H_C(f) = H_L + H_D(f); \quad (2)$$

where H_L is the path loss of a LoS channel; and $H_D(f)$ is the channel gain caused by the multi-path propagation and the diffuse effect. Specifically, the LoS channel fading is defined as follows [20]:

$$H_L = \begin{cases} \frac{(m+1)A_p g(\theta) T_s(\theta)}{2\pi(z^2 + w^2)} \cos^m(\phi) \cos(\theta), & 0 \leq \theta \leq \Theta_F \\ 0, & \theta > \Theta_F \end{cases} \quad (3)$$

where m is the Lambertian index which is a function of the half-intensity radiation angle $\theta_{1/2}$, expressed as $m = -1/\log_2(\cos(\theta_{1/2}))$; A_p is the physical area of the PD; z is the horizontal distance from a LiFi AP to the optical receiver; w is the height of the room; ϕ is the angle of irradiation; θ is the angle of incidence; Θ_F is the half angle of FoV of the receivers; $T_s(\theta)$ is the gain of the optical filter; and the concentrator gain $g(\theta)$ can be written as follows:

$$g(\theta) = \begin{cases} \frac{\chi^2}{\sin^2 \Theta_F}, & 0 \leq \theta \leq \Theta_F \\ 0, & \theta > \Theta_F, \end{cases} \quad (4)$$

where χ is the refractive index. The diffuse component is defined as follows [21]:

$$H_D(f) = \frac{\rho A_p e^{j2\pi f \Delta T}}{A_r(1-\rho)(1+j\frac{f}{f_c})}, \quad (5)$$

where A_r is the area of the room surface; ρ is the reflectivity of the walls; ΔT is the delay between the LoS signal and the onset of the diffuse signals; and f_c is the cut-off frequency of the optical channel. It is shown in [22] that this cut-off frequency is approximately 30 MHz in an indoor scenario. The front-end device frequency response is modelled as follows [23]–[25]:

$$H_F(f) = \exp\left(-\frac{f}{v f_0}\right); \quad (6)$$

where $v = 1.44$ is the fitting coefficient [26]; and f_0 is the 3 dB cut-off frequency of the front-end filtering effect. In general, the value of f_0 varies based on the characteristics of LEDs. For off-the-shelf LEDs, the 3 dB bandwidth is around 5 MHz while it is shown in [2] that a μ LED can achieve more than 60 MHz of 3 dB bandwidth. In this study, a median value of this parameter is used, which is $f_0 = 30$ MHz. For simplicity, the shadowing effect induced by objects is not considered in this study.

In a LiFi system, the LED lamps operate in the linear region where the output optical power is proportional to the modulated input voltage. Here, this region is termed as the linear working region of the LED. The signals outside this region are clipped before transmission. The average electric power of signals and the average optical power takes the following relationship [27]:

$$\iota = P_{\text{opt}}/\sqrt{P_t}, \quad (7)$$

where ι is the power ratio; P_{opt} is the average transmitted optical power of a LiFi AP which is proportional to the DC bias; and P_t is the electric power of the signals without DC bias. It is shown in [19] that an increase of ι results in a decrease of the probability of LiFi signals outside the LED linear working region. In general, $\iota = 3$ means that approximately 0.3% of the signals are clipped. In this case, the clipping noise can be considered negligible.

The signal-to-interference-plus-noise ratio (SINR) for user μ connected to LiFi AP α can be written as follows:

$$\text{SINR}_{\mu,\alpha}(f) = \frac{(\kappa P_{\text{opt}} H_{\mu,\alpha}(f))^2}{\iota^2 N_L B_L + \sum (\kappa P_{\text{opt}} H_{\mu,\text{else}}(f))^2}, \quad (8)$$

where κ is the optical to electric conversion efficiency at the receivers; N_L is the noise power spectral density in LiFi link; B_L is the baseband modulation bandwidth for each LiFi AP; $H_{\mu,\alpha}(f)$ is the channel gain in the frequency domain between user μ and LiFi AP; and $H_{\mu,\text{else}}(f)$ is the channel gain between user μ and the interfering LiFi APs, according to Eq. (1).

C. RF Channel Model

The RF system operates in the spectrum between 2.4 GHz and 2.4835 GHz [28]. In order to avoid ICI, this spectrum for

downlink transmission is divided into several non-overlap sub-channels, and each RF AP utilises one sub-channel for data transmission. Also, the OFDM technique is used to handle the multi-path propagation effect by a single-tap equaliser. In an indoor scenario, the RF channel gain is affected by the path-loss and multipath fading which is modelled by [29]:

$$G_{\mu,\alpha}(f) = \sqrt{10^{-\frac{L(d)}{10}}} h_r, \quad (9)$$

where $G_{\mu,\alpha}(f)$ is the frequency response at frequency, f , of the fading channel gain between user μ and RF AP α ; and d is the distance between user μ and RF AP α . Here h_r is modelled by a Rayleigh variate on each OFDM sub-carrier with variance 2.46 dB [29]; and $L(d)$ is the large-scale fading loss in decibels at the separation distance, d . The path loss, $L(d)$ is given by [28]:

$$L(d) = L(d_0) + 10\nu \log_{10}(d/d_0) + X, \quad (10)$$

where $L(d_0) = 47.9$ dB is the reference path loss at some reference distance $d_0 = 1$ m; $\nu = 1.6$ is the path loss exponent; and X is the shadowing component which is assumed to be a zero mean Gaussian distributed random variable with a standard deviation of 1.8 dB. Since there is no ICI in the RF system, the SINR is equivalent to SNR on each sub-carrier, which can be expressed as follows:

$$\text{SNR}_{\mu,\alpha}(f) = \frac{|G_{\mu,\alpha}(f)|^2 \Delta P_R}{N_R \Delta B_R}, \quad (11)$$

where ΔB_R is the modulation bandwidth of each sub-carrier; ΔP_R is the transmit power allocated to each sub-carrier; and N_R is the noise power spectral density in RF link. It is assumed that the transmit power is allocated equally to each sub-carrier. Thus, it is obtained that $\Delta P_R / \Delta B_R = P_R / B_R$, where P_R and B_R are the total transmit power and modulation bandwidth for a RF AP.

D. Adaptive Modulation Scheme

The number of OFDM sub-carriers used in LiFi and RF is denoted by Q . In order to improve the spectrum efficiency, adaptive bit loading is employed over the OFDM sub-carriers [19]. Specifically, M -ary quadrature amplitude modulation (M -QAM) ($M = 2^q$, $q = 1, 2, 3, \dots$) is considered. When $q = 1$ is achieved, M -QAM is converted to on-off keying (OOK) modulation. For simplicity, the required SINR (or SNR) for M -QAM can be expressed as a function of M , which follows [30], [31]:

$$\text{SINR (or SNR)} \approx -\frac{1}{3}(M-1) \ln(5P_e), \quad (12)$$

where M denotes the constellation size; and P_e is the bit error rate (BER) per sub-carrier. In this study, the BER is set to be 10^{-3} , where the channel coding is not considered. Since the baseband bandwidth in the LiFi system is B_L , its OFDM bandwidth is $2B_L$. The link communication data rate of LiFi between user μ and AP α can be written as follows [2]:

$$Z_{\mu,\alpha} = \frac{2B_L}{Q} \sum_{i=0}^{\frac{Q}{2}-1} q_L(i), \quad (13)$$

where $q_L(i)$ is the spectral efficiency on sub-carrier i in the LiFi system, according to Eq. (8) and Eq. (12). The link communication data rate of RF can be written as follows:

$$\Upsilon_{\mu,\alpha} = \frac{B_R}{Q} \sum_{i=0}^{Q-1} q_R(i), \quad (14)$$

where $q_R(i)$ is the spectral efficiency on sub-carrier i in the RF system, according to Eq. (11) and Eq. (12).

III. DYNAMIC LOAD BALANCING SCHEME

In this section, a dynamic load balancing scheme incorporating handover overhead is proposed. In general, the handover time in an indoor scenario is in the order of millisecond (ms) [10], and it is often modelled as a Poisson random process [32]. The probability mass function (PMF) of handover time is given by:

$$\Pr(t_{ij} = x) = \frac{\zeta_{ij}^x e^{-\zeta_{ij}}}{x!}, \quad x = 0, 1, 2, \dots \quad (15)$$

where t_{ij} is the overhead during the switch from AP i to AP j ; and $\zeta_{ij} = \mathbb{E}[t_{ij}]$ is the expectation of the overhead. The handover overhead causes a certain reduction in the average data rate between the serving AP and the user. In this study, the handover efficiency between two consecutive states can be written as follows:

$$\eta_{ij} = \begin{cases} \left[1 - \frac{t_{ij}}{T_p}\right]^+, & i \neq j \\ 1, & i = j, \end{cases} \quad i, j \in C_L \cup C_R. \quad (16)$$

where the operation $[\cdot]^+$ represents $\max(\cdot, 0)$. The link data rate between AP α and user μ in state n with handover considered can be expressed as follows:

$$r_{\mu,\alpha}^{(n)} = \begin{cases} \eta_{\alpha'\alpha} Z_{\mu,\alpha}^{(n)}, & \alpha \in C_L \\ \eta_{\alpha'\alpha} \Upsilon_{\mu,\alpha}^{(n)}, & \alpha \in C_R, \end{cases} \quad (17)$$

where α' is the AP allocated to user μ in the state $n-1$; $\eta_{\alpha'\alpha}$ is the handover efficiency between state $n-1$ and state n according to Eq. (16); and $Z_{\mu,\alpha}^{(n)}$ and $\Upsilon_{\mu,\alpha}^{(n)}$ are the data rates of the LiFi link and the RF link, respectively, according to Eq. (13) and Eq. (14). The value of $r_{\mu,\alpha}^{(n)}$ between user μ and AP α is calculated by the CU in each state.

Since the network load balancing consists of the APA and the RA, two variables $g_{\mu,\alpha}^{(n)}$ and $k_{\mu,\alpha}^{(n)}$ are introduced to model the load balancing in state n . The variable $g_{\mu,\alpha}^{(n)}$ is a binary number which equals 1 when user μ is connected to AP α , and otherwise it is 0. The variable $k_{\mu,\alpha}^{(n)}$ is a fractional number between 0 and 1, which represents the probability that user μ is served by AP α in state n . Statistically, $k_{\mu,\alpha}^{(n)}$ is equivalent to the proportion of the time resource allocated to user μ in each signal frame. The dynamic algorithm executed by the CU in N_s states is shown in Algorithm 1, and two load balancing algorithms that jointly and separately optimise the APA and the RA in each state are used by the CU. These two algorithms are given in Section IV and V, respectively.

IV. JOINT OPTIMISATION ALGORITHM (JOA)

In this section, the JOA that jointly optimises the AP assignment and the time resource allocation in each state is proposed. At first, the joint optimisation problem is formulated and an iterative algorithm for the problem is given. After that, the convergence and optimality of the proposed algorithm are analysed. Since the network load balancing of a specific state is studied, the superscript (n) in parameters $g_{\mu,a}^{(n)}$, $k_{\mu,a}^{(n)}$ and $r_{\mu,a}^{(n)}$ is omitted to enhance clarity.

A. Problem Formulation and Iterative Algorithm

When considering load balancing, it is crucial to apply an appropriate utility function. In some published papers [8], [9], [33], the β -proportional fairness function that considers both sum-rate and user fairness is used, which is defined in [34]:

$$\psi_\beta(x) = \begin{cases} \log(x), & \beta = 1 \\ x^{1-\beta}, & \beta \geq 0, \beta \neq 1, \end{cases} \quad (18)$$

where x is the achievable data rate; and β is the fairness coefficient. It is shown that the utility function $\psi_\beta(x)$ is monotonically increasing, strictly concave and continuously differentiable. The utility function in Eq. (18) includes several well known fairness concepts [33]. For instance, when $\beta = 1$, a proportional fairness is achieved, where users served by a specific AP share an equal proportion of time resource; and when $\beta \rightarrow \infty$, a max-min fair scheduler is realised. Particularly, when $\beta = 0$, a linear utility function is obtained which achieves a maximal system throughput. In this case, each AP only serves the user of the best CSI whereas the other users have zero data rates, which finally leads to an ineffective load balancing solution. Thus, the situation of $\beta = 0$ is beyond the scope of this study.

Using the utility function in Eq. (18), the joint optimisation of the APA and the RA can be formulated as follows:

$$\begin{aligned} & \max_{g_{\mu,a}, k_{\mu,a}} \sum_{\mu \in \mathcal{U}} \sum_{\alpha \in C_L \cup C_R} g_{\mu,a} \psi_\beta(k_{\mu,a} r_{\mu,a}) \\ & s.t. \sum_{\mu \in \mathcal{U}} g_{\mu,a} k_{\mu,a} \leq 1 \quad \forall \alpha \in C_L \cup C_R; \\ & \sum_{\alpha \in C_L \cup C_R} g_{\mu,a} = 1 \quad \forall \mu \in \mathcal{U}; \\ & g_{\mu,a} \in \{0, 1\}, \quad k_{\mu,a} \in [0, 1], \quad \forall \mu \in \mathcal{U}, \quad \forall \alpha \in C_L \cup C_R, \end{aligned} \quad (19)$$

where $r_{\mu,a}$ is the communication link data rate given in Eq. (17), which is a positive number. The optimum $k_{\mu,a}$ is shown to be greater than zero in Eq. (29) so that $\log(0)$ is avoided.

This optimisation problem in Eq. (19) is a mixed integer non-linear programming (MINLP) problem which is mathematically intractable because it involves both binary variables and real-valued positive variables. In order to simplify this problem, the binary variable $g_{\mu,a}$ is assumed to be a fractional number between 0 and 1. As shown in Appendix, the original problem is able to be converted into a concave optimisation problem using the fractional variable $g_{\mu,a}$. The Lagrangian

Algorithm 1 Dynamic Load Balancing Algorithm Executed by the CU

- 1: Initialisation: denote α'_μ as the AP allocated to user μ in state $n - 1$, $n \leftarrow 1$;
- 2: **while** $n \leq N_s$ **do**
- 3: Obtain the LiFi and RF CSI between each μ and α .
- 4: Calculate $R_{\mu,a}^{(n)}$ and $\Upsilon_{\mu,a}^{(n)}$ according to the CSI.
- 5: Calculate $\eta_{\alpha'_\mu a}^{(n)}$ from AP α'_μ to AP α for user μ .
- 6: Calculate $g_{\mu,a}^{(n)}$ and $k_{\mu,a}^{(n)}$ using $R_{\mu,a}^{(n)}$, $\Upsilon_{\mu,a}^{(n)}$ and $\eta_{\alpha'_\mu a}^{(n)}$, according to load balancing algorithm used in each state.
- 7: Restore the AP allocated to user μ in state n as α'_μ .
- 8: $n \leftarrow n + 1$;
- 9: **end while**

Multiplier method can therefore be used to solve this optimisation problem [9]. The Lagrangian function is expressed as follows:

$$L(g_{\mu,a}, k_{\mu,a}, \omega_a) = \sum_{\mu \in \mathcal{U}} \sum_{\alpha \in C_L \cup C_R} g_{\mu,a} [\psi_\beta(k_{\mu,a} r_{\mu,a}) - \omega_a k_{\mu,a}] + \sum_{\alpha \in C_L \cup C_R} \omega_a, \quad (21)$$

where $g_{\mu,a} \in [0, 1]$, $k_{\mu,a} \in [0, 1]$ and $\sum_{\alpha \in C_L \cup C_R} g_{\mu,a} = 1$. The Lagrangian multiplier ω_a corresponds to the α -th constraints in Eq. (20). The optimum of $g_{\mu,a}$ and $k_{\mu,a}$ can be obtained by solving the problem:

$$\min_{\omega_a} \max_{g_{\mu,a}, k_{\mu,a}} L(g_{\mu,a}, k_{\mu,a}, \omega_a). \quad (22)$$

In this study, the problem in Eq. (22) is solved by a distributed algorithm via Lagrangian decomposition [35]. According to the Lagrangian function in Eq. (21), a dual objective function in terms of user μ is defined as:

$$\Omega_\mu(g_{\mu,a}, k_{\mu,a}) = \sum_{\alpha \in C_L \cup C_R} g_{\mu,a} [\psi_\beta(k_{\mu,a} r_{\mu,a}) - \omega_a k_{\mu,a}]. \quad (23)$$

The problem in Eq. (22) is separated into two steps of optimisation. In the first step, the dual objective function is optimised, and this sub-problem can be formulated as follows:

$$\{g_{\mu,a}^*, k_{\mu,a}^*\} = \arg \max_{g_{\mu,a}, k_{\mu,a} \in [0, 1]} \Omega_\mu(g_{\mu,a}, k_{\mu,a}). \quad (24)$$

It can be seen that the function $\Omega_\mu(g_{\mu,a}, k_{\mu,a})$ is concave with respect to the variables $g_{\mu,a}$ and $k_{\mu,a}$. The partial derivatives of Ω_μ with $g_{\mu,a}$ and $k_{\mu,a}$ are:

$$\frac{\partial \Omega_\mu}{\partial g_{\mu,a}} = \psi_\beta(k_{\mu,a} r_{\mu,a}) - \omega_a k_{\mu,a}, \quad (25)$$

$$\frac{\partial \Omega_\mu}{\partial k_{\mu,a}} = g_{\mu,a} \left[\frac{\partial \psi_\beta(k_{\mu,a} r_{\mu,a})}{\partial k_{\mu,a}} - \omega_a \right], \quad (26)$$

where $\psi_\beta(\cdot)$ is given in Eq. (18) and its derivative with respect to $k_{\mu,a}$ is:

$$\frac{\partial \psi_\beta(k_{\mu,a} r_{\mu,a})}{\partial k_{\mu,a}} = \frac{r_{\mu,a}}{(k_{\mu,a} r_{\mu,a})^\beta}. \quad (27)$$

In conjunction with Eq. (27), it can be shown that Eq. (26) monotonically decreases as $k_{\mu,a}$ increases. Also, due to the strict positivity of $k_{\mu,a}, r_{\mu,a}$ and $\beta \neq 0$ it can be derived that:

$$\lim_{k_{\mu,a} \rightarrow 0^+} \frac{r_{\mu,a}}{(k_{\mu,a} r_{\mu,a})^\beta} = +\infty. \quad (28)$$

Since ω_a is finite and $g_{\mu,a}$ is non-negative, it can be seen that the condition, $\frac{\partial \Omega_\mu}{\partial k_{\mu,a}}|_{k_{\mu,a}=0} \geq 0$ is satisfied. Thus, the optimum $k_{\mu,a}$ can be calculated by:

$$k_{\mu,a}^* = \begin{cases} 1, & \frac{\partial \Omega_\mu}{\partial k_{\mu,a}}|_{k_{\mu,a}=1} \geq 0 \\ \frac{1}{r_{\mu,a}} \left(\frac{r_{\mu,a}}{\omega_a} \right)^{\frac{1}{\beta}}, & \frac{\partial \Omega_\mu}{\partial k_{\mu,a}}|_{k_{\mu,a}=1} < 0. \end{cases} \quad (29)$$

With $k_{\mu,a}^*$, the optimum $g_{\mu,a}$ is expressed as follows:

$$g_{\mu,a}^* = \begin{cases} 1, & \alpha = \arg \max_{a' \in C_L \cup C_R} \frac{\partial \Omega_\mu(k_{\mu,a'}^*)}{\partial g_{\mu,a'}} \\ 0, & \text{Otherwise.} \end{cases} \quad (30)$$

In the second step, the dual optimum $\{g_{\mu,a}^*, k_{\mu,a}^*\}$ is introduced to the optimisation problem in Eq. (22), and this problem can be converted to:

$$\min_{\omega_a} L(\Omega_\mu(g_{\mu,a}^*, k_{\mu,a}^*), \omega_a), \quad (31)$$

where

$$L(\Omega_\mu(g_{\mu,a}^*, k_{\mu,a}^*), \omega_a) = \sum_{\mu \in \mathcal{U}} \Omega_\mu(g_{\mu,a}^*, k_{\mu,a}^*) + \sum_{\alpha \in C_L \cup C_R} \omega_\alpha. \quad (32)$$

The objective function in Eq. (31) is concave and differentiable. Thus it can be solved by using the gradient projection method [36]. The update of ω_a is expressed as follows:

$$\omega_a(t+1) = \left[\omega_a(t) - \varepsilon_\omega \left(1 - \sum_{\mu \in \mathcal{U}} g_{\mu,a}^* k_{\mu,a}^* \right) \right]^+, \quad (33)$$

where ε_ω is the step size taken in the direction of the negative gradient of ω_a .

Based on the analysis above, the JOA can be regarded as a distributed iterative algorithm summarised in Algorithm 2, where $L(t)$ is the objective function in Eq. (22) in the t -th iteration; and ε is a threshold which is used to determine whether the algorithm converges.

B. Convergence Analysis of JOA

Typically, JOA converges after several iterations. According to Eq. (33), the step size ε_ω can significantly affect the speed of convergence. In general, a larger value of ε_ω can result in a faster convergence. However, when ε_ω is set to be very large, the algorithm becomes unstable so that the objective function tends to oscillate and the convergence cannot be guaranteed. In order to facilitate both the speed and the stability of the

Algorithm 2 JOA in Each State

```

1: Initialisation:  $\omega_a(t)$ ,  $\varepsilon_\omega$  and  $\varepsilon$ ;  $t \leftarrow 0$ ,  $L(t+1) \rightarrow \infty$ ,  $L(t) = 0$ .
2: while  $|L(t+1) - L(t)| > \varepsilon$  do
3:   for all each user  $\mu \in \mathcal{U}$  and each AP  $\alpha \in C_L \cup C_R$  do
4:     The CU calculates  $k_{\mu,a}^*$  according to Eq. (29).
5:     The CU calculates  $g_{\mu,a}^*$  according to Eq. (30).
6:   end for
7:   The CU broadcasts  $g_{\mu,a}^*$  and  $k_{\mu,a}^*$  to all of the users and APs.
8:   for all each AP  $\alpha \in C_L \cup C_R$  do
9:     AP  $\alpha$  updates the Lagrangian multiplier  $\omega_a$  by using  $g_{\mu,a}^*$  and  $k_{\mu,a}^*$ , according to Eq. (33).
10:    AP  $\alpha$  broadcasts the updated  $\omega_a$ .
11:   end for
12:   The CU calculates the objective function  $L(t)$  by using  $g_{\mu,a}^*$ ,  $k_{\mu,a}^*$  and  $\omega_a$ , according to Eq. (21).
13:    $t \leftarrow t + 1$ ;
14: end while
15: Output:  $g_{\mu,a}^*$  and  $k_{\mu,a}^*$ ;

```

convergence, an adaptive step size is considered and designed as follows:

$$\varepsilon_\omega \leftarrow \varepsilon_\omega d^{\frac{t}{2}}, \quad (0 < d < 1) \quad (34)$$

where t is the number of iterations and d is a bias weight. This step size decreases with t , which enables a more efficient iterations. Moreover, the step size, ε_ω approaches zero when t is large, and its gradient is affected by d . It is possible that ε_ω tends to zero before the algorithm converges. Hence, d should be well designed. In this study, it is set to be between 0.8 and 0.9. The simulations confirm that this range is a suitable choice for the simulation scenario considered in this paper.

Since the threshold, ε in JOA cannot be zero, the optimum $k_{\mu,a}^*$ gained in JOA may not strictly satisfy the constraint in Eq. (20). In order to avoid the situation $\sum_{\mu \in \mathcal{U}} g_{\mu,a}^* k_{\mu,a}^* > 1$, the optimum $k_{\mu,a}^*$ achieved by Algorithm 2 should be normalised, which can mathematically be expressed as follows:

$$\bar{k}_{\mu,a} = \frac{k_{\mu,a}^*}{\sum_{\mu \in \mathcal{U}} g_{\mu,a}^* k_{\mu,a}^*}, \quad \forall \alpha \in C_L \cup C_R, \quad (35)$$

where $g_{\mu,a}^*$ is the optimum of the variable $g_{\mu,a}$ obtained by using JOA. The normalised resource proportion $\bar{k}_{\mu,a}$ strictly satisfies the constraint in Eq. (20). The load balancing results achieved by JOA can be denoted as follows:

$$\begin{cases} g_{\mu,a}^{(\text{JOA})} &= g_{\mu,a}^* \\ k_{\mu,a}^{(\text{JOA})} &= \bar{k}_{\mu,a}. \end{cases} \quad (36)$$

C. Optimality Analysis

Due to the dual decomposition method used in JOA, it is difficult to discover whether the results attained in Eq. (36) after convergence are the global optima or not. Since a closed-form solution of the original optimisation problem is

mathematically intractable, an exhaustive search is used to find the global optimum.

With the assumption of N_u users and $N_l + N_r$ APs in this network, there are $(N_l + N_r)^{N_u}$ possibilities of AP allocation. Note that the AP assignments are denoted as $\mathbf{g}^{(i)}$, $i = 1, 2, \dots, (N_l + N_r)^{N_u}$. When a certain $\mathbf{g}^{(i)}$ is considered, the joint optimisation problem in Eq. (19) can be transformed as follows:

$$\max_{k_{\mu,a}^{(i)}} \sum_{a \in C_L \cup C_R} \sum_{\mu \in \mathcal{U}_a} \psi_{\beta}(k_{\mu,a}^{(i)} r_{\mu,a}), \quad (37)$$

where $\mathcal{U}_a$ is the set of users allocated to AP a . According to Eq. (37), the optimisation of $k_{\mu,a}$ can be undertaken independently in each cell, which is expressed as follows:

$$\max_{k_{\mu,a}^{(i)}} \sum_{\mu \in \mathcal{U}_a} \psi_{\beta}(k_{\mu,a}^{(i)} r_{\mu,a}), \quad (38)$$

$$s.t. \sum_{\mu \in \mathcal{U}_a} k_{\mu,a}^{(i)} \leq 1, \quad (39)$$

The Lagrangian multiplier method is used to solve this problem. The Lagrangian function is given by:

$$F(k_{\mu,a}^{(i)}, \omega) = \sum_{\mu \in \mathcal{U}_a} \psi_{\beta}(k_{\mu,a}^{(i)} r_{\mu,a}) + \omega \left(1 - \sum_{\mu \in \mathcal{U}_a} k_{\mu,a}^{(i)} \right), \quad (40)$$

where ω is the Lagrangian multiplier for the constraint in Eq. (39). The optimum $k_{\mu,a}^{(i)}$ can be found by making the gradient of the Lagrangian function in Eq. (40) equal to 0, which is written as follows:

$$\frac{\partial F(k_{\mu,a}^{(i)}, \omega)}{\partial k_{\mu,a}^{(i)}} = \frac{r_{\mu,a}}{(k_{\mu,a}^{(i)} r_{\mu,a})^{\beta}} - \omega = 0, \quad \mu \in \mathcal{U}_a. \quad (41)$$

It can be derived from Eq. (41) that:

$$k_{\mu,a}^{(i)} = \frac{r_{\mu,a}^{\frac{1}{\beta}-1}}{\omega^{\frac{1}{\beta}}}. \quad (42)$$

According to Eq. (42) and Eq. (39), it can be found that:

$$\omega^{\frac{1}{\beta}} = \sum_{\mu \in \mathcal{U}_a} r_{\mu,a}^{\frac{1}{\beta}-1}. \quad (43)$$

Combining Eq. (42) and Eq. (43), the optimum $k_{\mu,a}^{(i)}$ can be expressed as follows:

$$k_{\mu,a}^{(i)} = \frac{r_{\mu,a}^{\frac{1}{\beta}-1}}{\sum_{i \in \mathcal{U}_a} r_{i,a}^{\frac{1}{\beta}-1}} \quad (\beta > 0). \quad (44)$$

Given a certain AP allocation $\mathbf{g}^{(i)}$, it is possible to compute the objective function in Eq. (37) based on the RA result in Eq. (44). Subsequently, the global optimum of the original joint optimisation problem can be determined using exhaustive search. A comparison between the JOA and the global optimum is shown in Fig. 2. It can be seen that the JOA technique approaches the optimal solution very closely. Also, the gap between the JOA result and the global optimum diminishes with an increase in β . Therefore, it can be concluded that the JOA attains the performance of the global optimal solution to the original optimisation problem in Eq. (19).

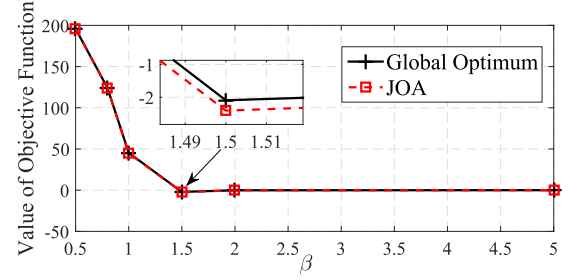


Fig. 2. Comparison of optimums between global optimisation and JOA (1000 times of independent and identical simulations are considered.)

V. SEPARATE OPTIMISATION ALGORITHM (SOA)

In this section, the SOA that separately optimises the AP assignment and the time resource allocation in each state is proposed. As before, the superscript n is omitted in parameters $g_{\mu,a}^{(n)}$, $k_{\mu,a}^{(n)}$ and $r_{\mu,a}^{(n)}$ for clarification.

A. AP Assignment in SOA

In order to make use of the high spatial spectrum efficiency of LiFi, the APA step in SOA can be realised as follows: users achieving LiFi data rates higher than a certain threshold, denoted by γ , are served by LiFi APs; and the others of data rates falling below the threshold via LiFi links would be assigned to the RF APs. Moreover, the criterion of maximal effective throughput is applied for the APA step. In terms of user μ , the LiFi AP offering the maximal link data rate involving handover is expressed as follows:

$$\tau_{1,\mu} = \arg \max_{j \in C_L} r_{\mu,j}, \quad (45)$$

where $r_{\mu,j}$ is the LiFi data rate according to Eq. (17). Assuming that the time resource is equally shared by users in each cell, the potential LiFi data rate of user μ can be written as follows:

$$\lambda_{\mu} = r_{\mu,\tau_{1,\mu}} / M_{\tau_{1,\mu}}, \quad (46)$$

where $M_{\tau_{1,\mu}}$ is the number of users served by LiFi AP $\tau_{1,\mu}$.

According to the principle of APA step, users following $\lambda_{\mu} < \gamma$ should be allocated to RF APs. The RF AP assigned to user μ can be written as follows:

$$\tau_{2,\mu} = \arg \max_{j \in C_R} r_{\mu,j}, \quad \lambda_{\mu} < \gamma. \quad (47)$$

Based on Eq. (45) (47), the APA result of SOA is expressed as follows:

$$g_{\mu,a}^{(\text{SOA})} = \begin{cases} 1, & a = \begin{cases} \tau_{1,\mu}, & \lambda_{\mu} \geq \gamma \\ \tau_{2,\mu}, & \lambda_{\mu} < \gamma \end{cases} \\ 0, & \text{Otherwise.} \end{cases} \quad (48)$$

B. Resource Allocation in SOA

In the RA step, each AP allocates the time resources to the connected users independently. Similar to JOA, the generalised utility function in Eq. (18) that considers both sum-rate and user fairness is used so that the RA can be formulated as

Algorithm 3 SOA in Each State

```

1: Initialisation:  $r_{\mu,\alpha}$  and  $\gamma$ .
2: for all user  $\mu$  do
3:   The CU calculates  $\tau_{1,\mu}$  according to Eq. (45).
4:   The CU calculates the optical data rate  $\lambda_{\mu}$  according to
     Eq. (46);
5:   if  $\lambda_{\mu} \geq \gamma$  then
6:     User  $\mu$  is allocated to LiFi AP  $\tau_{1,\mu}$ .
7:   else
8:     User  $\mu$  is allocated to RF AP  $\tau_{2,\mu}$ , according to Eq.
       (47).
9:   end if
10: end for
11: for all AP  $\alpha$  do
12:   Each AP determines the resource allocation for its serv-
     ing users, according to Eq. (49).
13: end for

```

a utility maximisation problem as expressed in Eq. (37). According to Eq. (44), the RA result of SOA consequently can be written as follows:

$$k_{\mu,\alpha}^{(\text{SOA})} = \frac{r_{\mu,\alpha}^{\frac{1}{\beta}-1}}{\sum_{i \in \mathcal{U}_{\alpha}} r_{i,\alpha}^{\frac{1}{\beta}-1}} \quad (\beta > 0). \quad (49)$$

Based on Eq. (48) and Eq. (49), the SOA can be summarised in Algorithm 3. In SOA, an optimal threshold is very crucial to enhance the system performance. It appears that a low threshold leads to overloaded LiFi attocells as well as an insufficient utilisation of RF resources whereas a high threshold leads to the inverse effect. In order to obtain an appropriate threshold, the characteristics of the LiFi and RF link data rates should be considered. These are closely related to the hybrid network topology and user distribution. Consequently, deriving a general closed-form solution of the optimal threshold in SOA is excessively complex to attain. In this study, therefore, a numerical simulation approach is used to find the best threshold.

VI. QoS ENHANCEMENT IN JOA AND SOA

In Section IV and Section V, two load balancing algorithms, JOA and SOA, are proposed. The user data rate in state n for both JOA and SOA can be written as follows:

$$R_{\mu}^{(n)} = \sum_{\alpha \in \mathcal{C}_L \cup \mathcal{C}_R} g_{\mu,\alpha}^{(n)} k_{\mu,\alpha}^{(n)} r_{\mu,\alpha}^{(n)}, \quad (50)$$

where $g_{\mu,\alpha}^{(n)}$ is given in Eq. (36) and Eq. (48) for JOA and SOA, respectively; and $k_{\mu,\alpha}^{(n)}$ is given in Eq. (36) and Eq. (49) for JOA and SOA, respectively. According to Eq. (50), the data rate of users that are using JOA can be expressed as a function of the fairness coefficient β , denoted by $R_{\mu,\text{JOA}}^{(n)}(\beta)$. Similarly, the data rate of users that are using SOA can be written as a function of the threshold γ and the fairness coefficient β , denoted by $R_{\mu,\text{SOA}}^{(n)}(\gamma, \beta)$. In JOA and SOA, the user data rate

can be significantly affected by β . An increase of β results in a decrease of the system sum-rate which, however, enables an improvement of the user fairness. Moreover, when employing SOA, the data rates are also affected by the threshold γ . The optimum threshold can contribute to an efficient utilisation of the multiple APs for users, which enables the system load to be well balanced.

In this study, a unified data rate request of users, denoted by Γ_0 , is considered in the hybrid network, which is regarded as the QoS of users. The outage probability of the user QoS is defined as follows:

$$\Phi_0 = \Pr(R_{\mu}^{(n)} < \Gamma_0). \quad (51)$$

This probability can be calculated by Monte Carlo simulations, which is expressed as follows:

$$\Phi_0 = \frac{\sum_n \text{Number of Users with } R_{\mu}^{(n)} < \Gamma_0}{\sum_n \text{Number of Total Users}} \quad (52)$$

In general, the design specifications of communication systems include an per user rate outage probability constraint. Let this outage probability constraint be denoted as Φ . The problem of maximising the QoS in JOA under the outage probability constraint can be formulated as follows:

$$\Gamma_{\text{JOA}} = \max_{\beta} \Gamma_0, \quad (53)$$

$$\text{s.t. } \Pr(R_{\mu,\text{JOA}}^{(n)}(\beta) < \Gamma_0) \leq \Phi, \quad (54)$$

and the problem formulation in SOA becomes:

$$\Gamma_{\text{SOA}} = \max_{\beta, \gamma} \Gamma_0, \quad (55)$$

$$\text{s.t. } \Pr(R_{\mu,\text{SOA}}^{(n)}(\beta, \gamma) < \Gamma_0) \leq \Phi, \quad (56)$$

where $R_{\mu,\text{JOA}}^{(n)}(\beta)$ and $R_{\mu,\text{SOA}}^{(n)}(\beta, \gamma)$ can be calculated according to Eq. (50). Since the analytical investigation of the problems in Eq. (53) and Eq. (55) is challenging, numerical simulations are used to find the optimum values for β and γ , respectively.

VII. PERFORMANCE EVALUATION AND DISCUSSION

A. Simulation Setup

As shown in Fig. 3, a 10 m $\times$ 5 m indoor office space is considered, which is equipped with 10 LiFi APs and 2 RF APs. It is assumed that the number of users in the simulation scenario is 30. All of the users move randomly in the indoor area, where the random way point model is applied [37]. The user speed is between 0 and 1 m/s. The average handover efficiency is defined as $\eta = \mathbb{E}[\eta_{ij}]$, where η_{ij} is defined in Eq. (16). The other parameters are given in Table I, which is based on published research [4], [5], [9], [13], [21]. In the simulation, the outage probability is calculated based on 2000 realisations. We focus on the outage probability ranging between 0 and 0.2, and disregard the values from 0.2 to 1 since they are not practical.

In Section VII.B, the computational complexity of JOA and SOA is analysed. In Section VII.C, the effects of β and γ on JOA and SOA are studied, and the optimum values of β and γ to maximise the QoS are obtained. In Section VII.D,

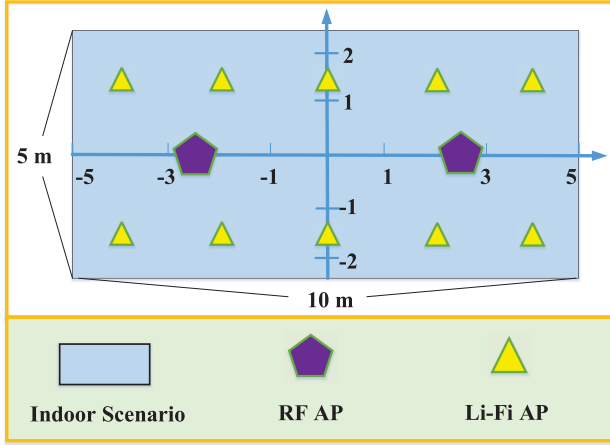


Fig. 3. Simulation scenario.

TABLE I
SIMULATION PARAMETERS

Name of Parameters	Value
Vertical distance between APs and users, w	2 m
Average transmitted optical power per LiFi AP, P_{opt}	9 W
Baseband modulation bandwidth for LED lamp, B_L	30 MHz
The physical area of a PD, A_p	1 cm ²
Half-intensity radiation angle, $\theta_{1/2}$	30 deg.
Gain of optical filter, $T_s(\theta)$	1.0
Receiver FoV semi-angle, Θ_F	90 deg.
Refractive index, χ	1.5
Optical to electric conversion efficiency, κ	0.53 A/W
OFDM sub-carrier number, Q	64
Transmitted power for each RF AP, P_R	10 dBm
Modulation bandwidth for each RF AP, B_R	10 MHz
Noise power spectral density in LiFi, N_L	-210 dBm/MHz
Noise power spectral density in RF, N_R	-75 dBm/MHz
Resource allocation interval of central unit, T_p	200 ms

the effect of handover efficiency η on network load balancing are investigated, and the performance of maximum achievable QoS (MAQ) in JOA and SOA is compared with that of conventional load balancing schemes.

B. Evaluation of Computational Complexity

In JOA, $g_{\mu,\alpha}^{(JOA)}$ and $k_{\mu,\alpha}^{(JOA)}$ can be obtained after several iterations, and the required number of the iterations is denoted by I . In Fig. 4, the objective function in Eq. (22) with respect to the number of iterations is shown. It can be seen that I increases with the fairness coefficient β . This signifies that when β tends to be infinite, JOA is not feasible due to the prohibitive complexity required. In order to avoid a large number of iterations, the values for β in the range $0 < \beta < 10$ is only considered for JOA in this study.

In SOA, $g_{\mu,\alpha}^{(SOA)}$ and $k_{\mu,\alpha}^{(SOA)}$ can be calculated without iterations. The number of additions, multiplications and exponential operations in both JOA and SOA are summarised in Table. II, where $N_{ap} = N_l + N_r$, is the number of the APs. As shown, the number of the three types of computations in JOA are all greater than those in SOA. With the configuration $N_u = 30$, $N_{ap} = 12$ and $I = 40$, the average numbers of the three operations are evaluated in Matlab. The simulation

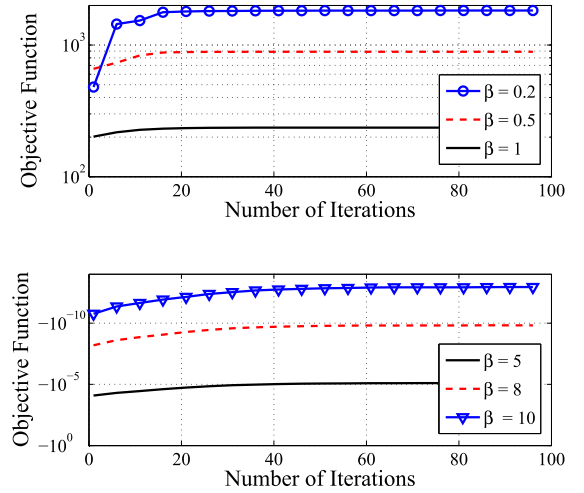
Fig. 4. The number of iterations with different fairness coefficient β .

TABLE II
COMPUTATION COMPLEXITY BETWEEN JOA AND SOA

Algorithm	Addition	Multiplication	Exponentiation
JOA	$O(N_u N_{ap} I)$	$O(N_u N_{ap} I)$	$O(N_u N_{ap} I)$
SOA	$O(N_u N_{ap})$	$O(N_u)$	$O(N_u)$

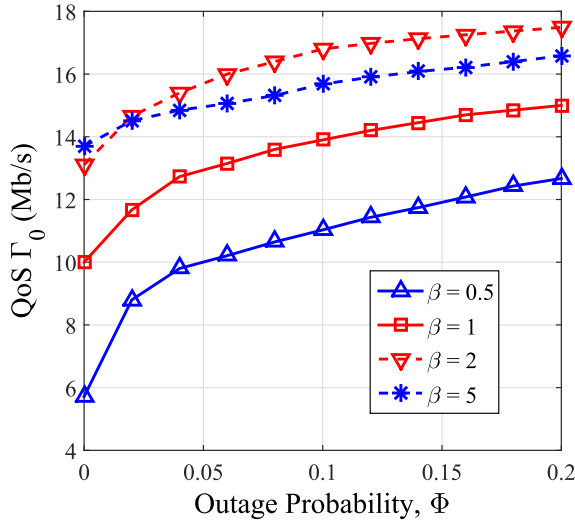
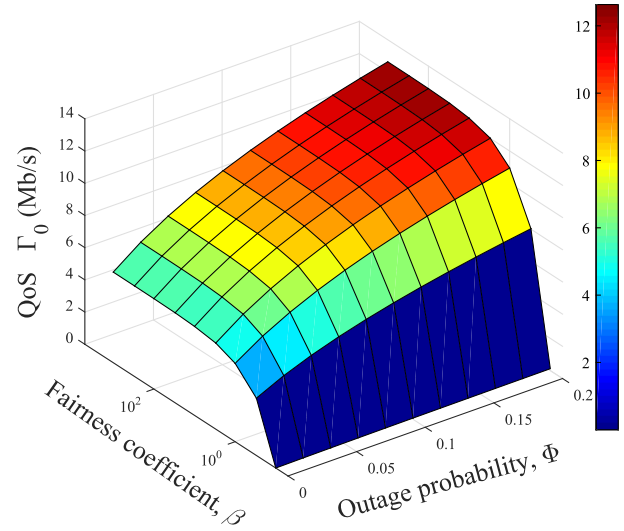
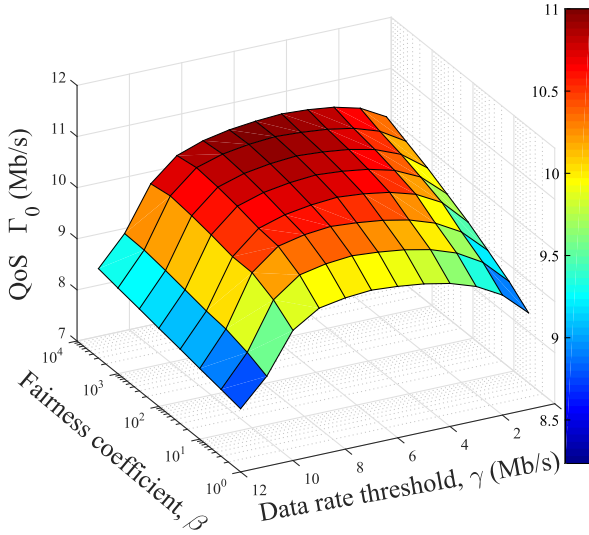
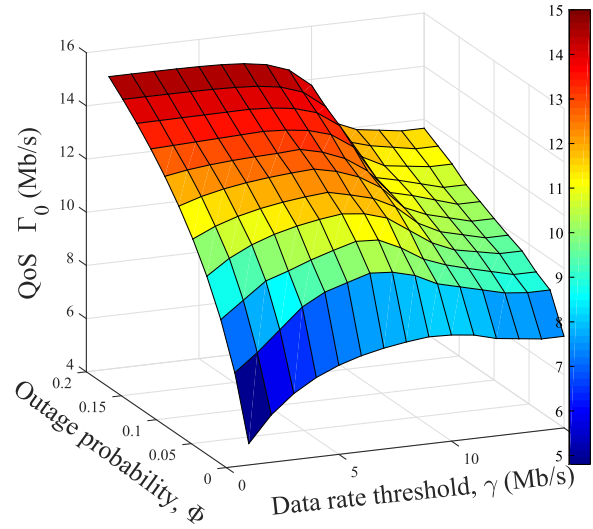
shows that the number of additions in JOA is 66 times greater than that in SOA; the number of multiplications in JOA is 1445 times greater than that in SOA; and the number of exponentiation in JOA is 1440 times greater than that in SOA. The gap between the computational complexity of JOA and SOA increases as the parameters, N_u , N_{ap} and I increase. Therefore, it can be concluded that SOA achieves much lower computational complexity than JOA.

C. Effects of β and γ on JOA and SOA

In this subsection, the maximum QoS achieved by the proposed schemes is analysed and the effects of β and γ on JOA and SOA are evaluated. The handover efficiency, η is set to be 1, which means the handover overhead is neglected in this case.

1) *JOA Case:* Fig. 5 shows that the QoS for JOA as a function of the outage probability. It can be seen that Γ_0 increases as β increases from 0.5 to 2, but reduces when β increases up to 5. This is because a higher β means better user fairness is achieved, resulting in an increase of Γ_0 . However, in order to improve the fairness performance, some time resource should be allocated to the users with poor channel gains, and this leads to a decrease in the user data rates. Thus, when β is set to be very large, Γ_0 in the hybrid network would be low. The simulation results show that the maximum QoS in JOA is achieved at $\beta = 2$ for $0.02 \leq \Phi \leq 0.2$, and $\beta = 5$ can achieve the maximum QoS for the range of $0 \leq \Phi < 0.02$.

2) *SOA Case:* The QoS performance for SOA is affected by both the threshold, γ and the fairness coefficient, β . Fig. 6 shows the QoS Γ_0 performance corresponding to β and γ with $\Phi = 0.1$. As shown, Γ_0 always increases with β for any threshold. Also, Fig. 7 shows that Γ_0 increases with β

Fig. 5. QoS Γ_0 with outage probability constraints in JOA ($\eta = 1$).Fig. 7. QoS Γ_0 with respect to β and Φ in SOA (γ is optimised, $\eta = 1$).Fig. 6. QoS Γ_0 with respect to β and threshold γ in SOA ($\Phi = 0.1$, $\eta = 1$).Fig. 8. QoS Γ_0 with respect to γ and Φ in SOA ($\beta \rightarrow +\infty$, $\eta = 1$).

when the outage probability constraint is between 0 and 0.2. It can be concluded that for given Φ and γ , QoS performance always improves as β increases. This means that the optimum fairness coefficient β tends to be positive infinite, where the max-min fairness is achieved. According to Eq. (29), the time resource proportion to achieve the maximum QoS in SOA can be written as follows:

$$k_{\mu,a}^{(\text{SOA})} = \frac{r_{\mu,a}^{-1}}{\sum_{i \in \mathcal{U}_a} r_{i,a}^{-1}}. \quad (57)$$

The time resource allocation in SOA is undertaken independently for each AP. Thus, β here represents the fairness of the users in each cell, and not in the entire system.

Fig. 8 shows the QoS of users with respect to γ and Φ , where β approaches infinity. It can be seen that Γ_0 is a concave function of γ for a specific, Φ . This is because a large threshold results in a large number of users allocated to RF APs, and thus the user data rates in the RF cells would be low.

On the other hand, for a small value of γ , the number of users served by LiFi increases so that a low data rate performance of LiFi is achieved. It can be seen that the optimal γ decreases as Φ increases. Specifically, the optimum γ is 9 Mb/s at $\Phi = 1\%$; and it decreases to 3 Mb/s at $\Phi = 15\%$. With the optimum γ and $\beta \rightarrow +\infty$, the maximum QoS in SOA can be achieved.

D. Effect of η on Load Balancing

The effect of handover efficiency η on JOA and SOA is evaluated and presented in Fig. 9. The maximum QoS of JOA and SOA with $\Phi = 0.1$ are 16.8 Mb/s and 12.8 Mb/s, respectively. This means that 90% of the users can approximately achieve 1.3 times higher data rate in JOA than in SOA. In both algorithms, the maximum QoS decreases along with η , and a large η results in a steep reduction. The simulation results show that when η increases from 0.25 to 1, the maximum QoS is improved approximately by 6.3 Mb/s and 5.6 Mb/s in JOA and SOA, respectively.

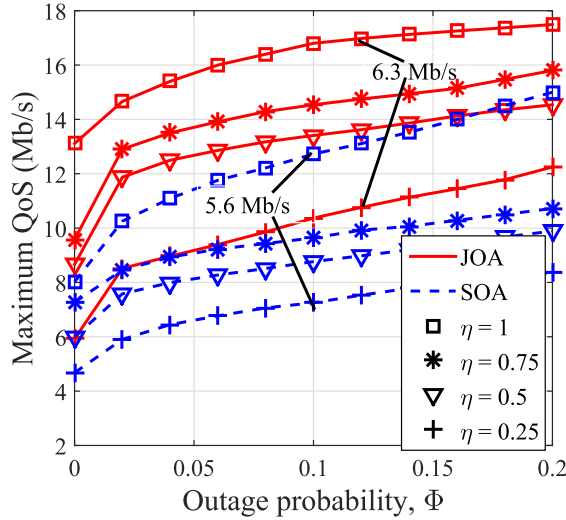


Fig. 9. The maximum QoS corresponding to Φ with different values of η by using JOA and SOA.

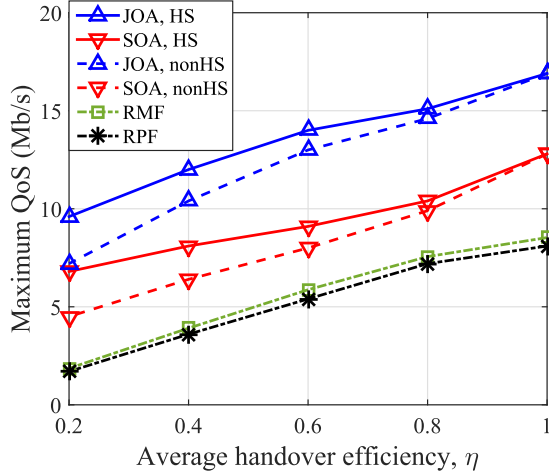


Fig. 10. The maximum QoS with the average handover efficiency for four different algorithms. ($\Phi = 0.1$).

Fig. 10 shows the performance of the maximum QoS for different algorithms. In the legend, ‘HS’ refers to the proposed handover scheme that is used in conjunction with load balancing algorithms while ‘nonHS’ refers to the direct handover strategy. It shows that the proposed handover scheme is able to enhance the user QoS. Specifically, the proposed handover algorithm achieves a data rate improvement of 2 Mb/s over the ‘nonHS’ scheme. Apart from JOA and SOA, two benchmark algorithms are considered for comparison, where the AP assignment and the resource allocation are undertaken separately. In the first benchmark algorithm, random AP allocation and max-min fairness scheduler are used, termed as ‘RMF’. In the second benchmark algorithm, random AP allocation and proportional fairness scheduler are used, termed as ‘RPF’. It is shown that the maximum QoS achieved by RMF and RPF are very close, and this maximum QoS in JOA and SOA are approximately 8 Mb/s and 3 Mb/s higher than that in RMF and RPF, respectively. Moreover, the performance of the

TABLE III
SYSTEM AVERAGE DATA RATE ($\Phi = 0.1$)

η	JOA (Mb/s)	SOA (Mb/s)	RMF (Mb/s)	RPF (Mb/s)
0.5	28.42	18.77	18.05	18.21
1	38.61	26.94	25.89	26.51

system average data rate for these four algorithms is evaluated and given in Table. III. It can be seen that the average data rate in SOA is slightly higher than that in RMF and RPF, and JOA outperforms the other three algorithms. For the the average data rate, JOA achieves an improvement of 50% compared to SOA.

A lower handover efficiency means that additional capacity would be sacrificed because of the increased handover overhead. Theoretically, it is possible that the handover time is longer than T_p and the handover efficiency is 0. In that case, users cannot select other APs because the handovers can not be finished during the current state. Thus, the achievable data rate during this state is zero if users try to switch APs. In order to implement the proposed load balancing schemes, the handover overhead between any two APs should be estimated in each state. The Kalman filtering approach can be applied for the overhead estimation [38]. Since it is outside the scope of this study, the overhead estimation is omitted here and will be the subject of future research.

VIII. CONCLUSION

In this study, a dynamic load balancing scheme for indoor LiFi and RF hybrid networks is proposed, and the handover overhead caused by user mobility is considered. Two load balancing algorithms, JOA and SOA, are proposed which jointly and separately optimise APA and RA, respectively. In order to achieve the maximum unified QoS, the optimum fairness coefficient and data rate threshold in both algorithms are obtained by using numerical simulations. Simulation results show that in a typical indoor scenario, the maximum QoS for more than 90% of the users in JOA is approximately 1.3 times higher than that in SOA. Also, the average data rate in JOA is 1.5 times higher than that in SOA. However, the computational complexity of JOA is significantly higher, and in some instances, it can be more than 1000 times greater than that of SOA. This means that SOA can offer a better performance/complexity trade-off than JOA for system load balancing.

APPENDIX

CONCAVITY PROOF OF JOINT OPTIMISATION PROBLEM

The concavity of the objective function in Eq. (19) with a fractional variable, $g_{\mu,\alpha}$ is proven in Appendix. It is assumed that $g_{\mu,\alpha}$ is a fractional number. The objective function in Eq. (19) is denoted as $F(g_{\mu,\alpha}, k_{\mu,\alpha})$, and its Hessian matrix can be written as follows:

$$\mathcal{H}(F) = \begin{bmatrix} \frac{\partial^2 F}{\partial g_{\mu,\alpha}^2} & \frac{\partial^2 F}{\partial g_{\mu,\alpha} \partial k_{\mu,\alpha}} \\ \frac{\partial^2 F}{\partial k_{\mu,\alpha} \partial g_{\mu,\alpha}} & \frac{\partial^2 F}{\partial k_{\mu,\alpha}^2} \end{bmatrix}, \quad (58)$$

where

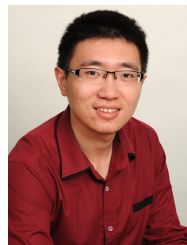
$$\frac{\partial^2 F}{\partial g_{\mu,a}^2} = 0; \quad (59)$$

$$\frac{\partial^2 F}{\partial k_{\mu,a}^2} = -\beta \frac{r_{\mu,a}^2}{(k_{\mu,a} r_{\mu,a})^{\beta+1}} \leq 0. \quad (60)$$

Since all of the principle minors of the Hessian matrix are non-positive, the objective function is concave with respect to the variables, $g_{\mu,a}$ and $k_{\mu,a}$.

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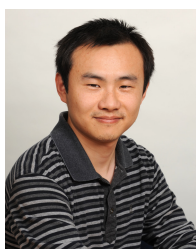


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